

Part X — Applied Physics · Chapter 53

Physics in Engineering

How the principles of physics become the tools, structures, and systems of the designed world

A bridge carries thousands of tonnes of traffic without bending. An aircraft wing generates lift from nothing but its shape moving through air. A power grid distributes electrical energy across continents at precisely 50 or 60 Hz. A smartphone contains an accelerometer, a gyroscope, a GPS receiver, a camera, and a radio transmitter, all squeezed into a rectangle thinner than a pencil. A hospital MRI machine wraps a superconducting electromagnet around a patient to image soft tissue without ionising radiation. None of these exist by accident: each is the deliberate application of physics — mechanics, thermodynamics, electromagnetism, wave physics, materials science — translated by engineers into systems that must work safely, reliably, and economically under real-world constraints. This chapter surveys how the physics studied throughout this course becomes engineering practice. It is not a new branch of physics; it is physics applied with purpose, discipline, and accountability for consequences.

53.1 What Engineering Does with Physics

Physics describes how the natural world works: it discovers laws, measures constants, and tests theories against experiment. Engineering uses those laws to build things: structures, machines, circuits, systems. The engineer's question is not "What does nature do?" but "How do we make nature do what we want?"

This shift from description to prescription changes the way physics is used. In physics, we often solve idealised problems: frictionless surfaces, ideal gases, point masses, perfect conductors. In engineering, friction matters, real gases deviate from ideal behaviour, masses have size and shape, and no conductor is perfect. Engineering requires not just applying the right formula, but choosing the right model, knowing when a simplification is justified, and understanding the consequences when a model fails.

Engineering disciplines map onto areas of physics. Civil and structural engineering draw heavily on statics, mechanics, and materials science. Mechanical engineering uses

dynamics, thermodynamics, fluid mechanics, and materials. Electrical engineering is built on electromagnetism, circuit theory, and quantum mechanics (for semiconductors). Aerospace engineering combines fluid mechanics, thermodynamics, structural mechanics, and control theory. Chemical engineering uses thermodynamics and reaction kinetics. Biomedical engineering applies mechanics, fluid dynamics, and electromagnetism to biological systems.

Engineering Discipline	Core Physics Areas	Typical Applications
Civil / Structural	Statics, mechanics, materials	Bridges, buildings, dams, roads
Mechanical	Dynamics, thermodynamics, fluids	Engines, machines, HVAC, vehicles
Electrical	Electromagnetism, circuits, QM	Power systems, electronics, communications
Aerospace	Fluid mechanics, thermodynamics	Aircraft, spacecraft, rockets
Chemical	Thermodynamics, kinetics	Refineries, reactors, materials processing
Biomedical	Mechanics, EM, fluid dynamics	Medical devices, imaging, prosthetics
Environmental	Fluid mechanics, thermodynamics	Water systems, air quality, energy

Table 53.1 *Mapping of engineering disciplines to their primary physics foundations.*

53.2 Design Constraints and Tradeoffs

Every engineering design exists within a set of constraints: requirements that limit what solutions are acceptable. Understanding, quantifying, and satisfying constraints is the core of engineering design. Physics defines many of these constraints.

Physical constraints are the most fundamental: the laws of thermodynamics set limits on efficiency; stress limits in materials set maximum loads; electrical breakdown strength sets maximum voltages; diffraction limits optical resolution. These cannot be negotiated.

Practical constraints include cost, manufacturability, weight, size, power consumption, and maintenance requirements. A material with twice the strength at ten times the cost may not be the right choice. An engine with 5% higher efficiency but 20% greater mass may lose net performance in a vehicle. Engineering design is fundamentally about managing tradeoffs.

The design process typically involves specifying requirements, generating candidate solutions, analysing each against the physical constraints and practical constraints, prototyping and testing the most promising solutions, and iterating. Physics analysis is the tool that prunes the enormous space of possible solutions to those that are physically feasible.

The Rule: Tradeoff principle: In engineering, improving one performance parameter usually worsens another. Stronger materials are often heavier or more expensive. Higher efficiency engines often require higher temperatures, which require more expensive materials. Lighter structures are often less stiff. Recognising these tradeoffs is a core engineering competence.

Example — Bridge design tradeoff A suspension bridge can span much longer distances than a beam bridge with the same material, but it requires expensive cables and anchorages. A truss bridge is stiffer but heavier. A cable-stayed bridge is a compromise. The choice depends on the required span, traffic load, site conditions, cost, and aesthetic requirements. There is no universally optimal bridge type.

53.3 Forces, Loads, and Structural Design

Every structure must support loads without failing. A load is any force or moment applied to a structure. Dead loads are permanent (the structure's own weight). Live loads are variable (people, vehicles, wind, snow). Dynamic loads vary with time and can cause fatigue or resonance.

For a structure to be in static equilibrium, the net force and net moment (torque) on every part must be zero. Structural engineers use these conditions — the same conditions introduced in Chapter 4 — to calculate internal forces in beams, columns, trusses, and cables. These internal forces are characterised as:

- Tension: a pulling force tending to elongate a member. Cables and rods in suspension bridges carry tension.
- Compression: a pushing force tending to shorten a member. Columns and arches carry compression.
- Shear: a force parallel to a surface, tending to cause sliding. Beams under transverse loads experience shear.

- Bending moment: a tendency to rotate a cross-section. Beams under loads experience bending moments that create tension on one face and compression on the other.

The relationship between internal forces and the resulting stresses and strains is given by the elastic modulus (Young's modulus): $\sigma = E\varepsilon$, where σ is stress (force per area), E is the material's stiffness, and ε is strain (fractional deformation). This is Hooke's law in the language of materials science (Chapter 11).

$$\sigma = F / A$$

$$\varepsilon = \Delta L / L_0$$

$$\sigma = E\varepsilon$$

Stress σ (Pa), strain ε (dimensionless), and Young's modulus E (Pa). E for steel ≈ 200 GPa; for aluminium ≈ 70 GPa; for concrete ≈ 30 GPa; for bone ≈ 20 GPa.

Example — Column under load A steel column (cross-sectional area 0.01 m^2 , $E = 200 \text{ GPa}$) supports a $5 \times 10^6 \text{ N}$ load. Stress: $\sigma = F/A = 5 \times 10^6 / 0.01 = 500 \text{ MPa}$. This is near the yield strength of mild steel ($\sim 250 \text{ MPa}$) — the column would yield. A structural engineer would either increase the area or use higher-strength steel (yield strength up to 700 MPa).

Bridge Note *The structural concepts in this section are direct applications of statics (Chapter 4), Hooke's Law and elasticity (Chapter 11), and the mechanical properties of materials covered in Chapter 12. Structural engineering does not invent new physics; it applies these principles to real geometry and real materials under real loads.*

53.4 Materials and Failure Prevention

Choosing the right material is inseparable from structural design. Materials fail when the stress on them exceeds their strength. The key material properties for structural engineering are:

Property	Definition	Relevance
Young's modulus (E)	Stiffness; stress/strain in elastic range	Controls deformation under load
Yield strength	Stress at which permanent deformation begins	Sets working load limit (with safety factor)
Ultimate tensile strength	Maximum stress before fracture in tension	Sets absolute strength limit
Toughness	Energy absorbed before fracture	Resistance to impact and crack propagation

Fatigue strength	Stress amplitude that causes failure after many cycles	Critical for rotating machinery and bridges
Creep resistance	Resistance to slow deformation under sustained load	Important at high temperatures (turbines, boilers)
Corrosion resistance	Resistance to chemical degradation	Affects lifespan in harsh environments

Table 53.2 *Key material properties for structural and mechanical engineering design.*

Failure modes are the specific ways a structure or component can cease to function. Common failure modes include:

- **Yielding:** the material permanently deforms. For ductile materials like steel, yielding is visible and progressive — it is a warning, not a sudden catastrophe. Engineers design structures to yield rather than fracture under extreme load (ductility is a safety feature).
- **Fracture:** the material breaks. Brittle materials (glass, cast iron, some ceramics) fracture without warning. Fracture mechanics predicts when a crack will grow unstably.
- **Fatigue:** repeated cyclic loading below the yield strength causes slow crack growth that eventually leads to fracture. Fatigue failure caused the de Havilland Comet crashes in 1954 and the Aloha Airlines fuselage failure in 1988. Engineers use fatigue analysis (S-N curves) to predict service life.
- **Buckling:** a slender column under compression fails by bending sideways suddenly, even before the material yields. Buckling is a geometric instability, not a material failure — a thin steel ruler buckles at a fraction of its compressive strength.

Historical Note: The Tacoma Narrows Bridge collapsed in 1940, just four months after opening, due to aerodynamic flutter — a resonance between its natural oscillation frequency and the shedding of vortices by wind. The bridge was not overloaded statically; it failed dynamically. The collapse prompted major advances in bridge aerodynamics and led to modern wind tunnel testing of bridge designs before construction.

53.5 Energy, Work, and Mechanical Systems

The energy principle — energy is conserved; it can be converted but not created or

destroyed — is the most powerful tool in engineering analysis. When a machine converts one form of energy to another, the efficiency η is the fraction of input energy that appears as useful output:

$$\eta = E_{\text{useful_output}} / E_{\text{input}} = W_{\text{out}} / W_{\text{in}}$$

Efficiency η (dimensionless; often expressed as a percentage). The remainder ($1 - \eta$) is wasted, usually as heat due to friction, viscosity, or electrical resistance. Real machines always have $\eta < 1$.

Thermodynamic efficiency limits are set by the laws of thermodynamics. For a heat engine operating between a high-temperature source T_H and a low-temperature sink T_C (in kelvin), the maximum possible efficiency is the Carnot efficiency (Chapter 29):

$$\eta_{\text{Carnot}} = 1 - T_C / T_H$$

Carnot efficiency. This is the maximum efficiency any heat engine can achieve between temperatures T_H and T_C . Real engines are always less efficient. To increase η_{Carnot} , raise T_H or lower T_C ; engineers pursue both strategies.

Raising operating temperature to improve efficiency is a persistent theme in engineering. Modern gas turbine engines operate at combustor temperatures above 1700°C — above the melting point of the turbine blade material — and achieve this through active cooling systems where air from the compressor flows through channels inside the blades, keeping the metal below its melting point.

Example — Power plant efficiency A coal power plant operates between $T_H = 600^\circ\text{C} = 873\text{ K}$ and $T_C = 35^\circ\text{C} = 308\text{ K}$. Carnot efficiency $= 1 - 308/873 = 0.647 = 64.7\%$. A real steam cycle achieves about 35–45% due to irreversibilities. Modern combined-cycle gas turbine plants achieve about 60% efficiency by using both a gas turbine cycle and a steam cycle in series.

Key Point: The work-energy theorem ($W_{\text{net}} = \Delta KE$, Chapter 6) and conservation of mechanical energy (Chapter 7) are the tools for analysing individual machine components. The laws of thermodynamics govern efficiency limits for systems involving heat. Both are always active simultaneously in real thermal machines.

53.6 Motion, Forces, and Machine Design

Machines convert forces and motions into other forces and motions. The ratio of output force to input force is the mechanical advantage (MA). For an ideal (lossless) machine, energy is conserved: a large force over a small distance equals a small force over a large distance. Real machines lose some energy to friction, so actual mechanical advantage is always less than ideal.

Simple machines (levers, pulleys, inclined planes, screws, wheels and axles) are the

fundamental building blocks of complex mechanisms. Every complex machine — a car engine, a printing press, a robotic arm — is a combination of simple machines. The principles of torque (Chapter 8) and power (Chapter 6) govern their design.

Rotating machinery introduces additional considerations. For rotating components, the relevant quantities are angular velocity ω , torque τ , and rotational power $P = \tau\omega$. A gear system converts rotational speed and torque: a gear pair with ratio N (output teeth / input teeth) multiplies torque by N and divides angular velocity by N , conserving power (in the ideal case).

$$\tau_{\text{out}} / \tau_{\text{in}} = N = n_{\text{out}} / n_{\text{in}} \quad \omega_{\text{out}} / \omega_{\text{in}} = 1/N$$

Gear ratio relationships. N = tooth count ratio (output/input). A reduction gear ($N > 1$) increases torque and decreases speed. An overdrive gear ($N < 1$) decreases torque and increases speed. Power: $P = \tau\omega$ is conserved in an ideal gear pair.

Example — Engine and gearbox A car engine produces 200 N·m of torque at 3000 rpm ($\omega = 314$ rad/s). In first gear, the overall gear ratio is 12:1 ($N = 12$ at the wheels). Torque at the wheels: $200 \times 12 = 2400$ N·m (minus friction losses). Angular velocity at the wheels: $314/12 = 26.2$ rad/s $\approx$ 250 rpm. Power: $P = 200 \times 314 \approx 62.8$ kW (conserved in ideal transmission).

Dynamics — the behaviour of systems under changing forces and accelerations — is critical for vibration, shock, and fatigue analysis. Newton's second law ($F = ma$, $\tau = I\alpha$) applies to both translational and rotational dynamics. A machine component that resonates at its natural frequency under a periodic forcing function can fail rapidly. Engineers calculate natural frequencies and design systems so that operating frequencies avoid resonances (or use damping to limit resonance amplitude).

53.7 Fluids in Engineering Systems

Fluid mechanics (Part VII, Chapters 35–38) is central to an enormous range of engineering systems. Fluid flows carry energy, transport heat, transmit forces, generate lift, and drive turbines. The key principles are continuity (conservation of mass in flow), Bernoulli's equation (energy conservation in flow), and momentum (Newton's second law applied to fluids).

Hydraulic systems use incompressible fluids (usually oil or water) to transmit forces and power. Pascal's principle (Chapter 35) underlies hydraulic brakes, lifts, presses, and actuators. The power in a hydraulic system is $P = \Delta Pq$, where ΔP is pressure difference and q is volumetric flow rate.

Pneumatic systems use compressed gas. Because gases are compressible, pneumatic

systems store energy in compressed form and are used where actuator forces are moderate and spring-like behaviour is acceptable (air tools, pneumatic cylinders, air suspension).

Aerodynamics is the branch of fluid mechanics dealing with air flows around objects. Bernoulli's principle explains lift qualitatively: an aerofoil is shaped so that air flows faster over the upper surface, creating lower pressure above and higher pressure below. The net upward pressure difference provides lift. Modern aerofoil analysis is more nuanced (inviscid potential flow, boundary layer theory, computational fluid dynamics), but Bernoulli's principle provides the correct physical intuition.

Pipe flow in water distribution, chemical plant piping, and HVAC systems requires analysis of pressure drop due to friction. The Hagen-Poiseuille equation for laminar flow and empirical friction factor correlations for turbulent flow (both derived from fluid mechanics principles) determine the required pipe diameter, pump power, and pressure ratings.

Example — Pump selection A water supply system requires $0.05 \text{ m}^3/\text{s}$ of flow through a pipe of length 500 m , diameter 0.1 m , with a 30 m elevation gain and pipe friction losses of 25 m head. Total head required: $30 + 25 = 55 \text{ m}$. Pump power: $P = \rho g Q h = 1000 \times 9.81 \times 0.05 \times 55 = 26.9 \text{ kW}$ (ideal). Real pump would require about $35\text{--}40 \text{ kW}$ to account for pump efficiency of $70\text{--}75\%$.

53.8 Heat Transfer and Thermal Design

Thermal engineering involves managing the flow of heat into, through, and out of systems. The three mechanisms of heat transfer — conduction, convection, and radiation (Chapter 30) — all appear in engineering systems, often simultaneously.

Thermal resistance is the engineering analogue of electrical resistance. Just as $V = IR$ relates voltage, current, and resistance, the heat flow rate $\dot{Q}$ through a conductor is $\dot{Q} = \Delta T / R_{\text{th}}$, where ΔT is the temperature difference and R_{th} is the thermal resistance. Thermal resistances in series add directly, just like electrical resistances in series. This allows complex thermal systems (insulated walls, heat exchangers, electronic cooling systems) to be analysed with simple circuit-like diagrams.

Electronics cooling is a major thermal engineering challenge. Microprocessors generate $50\text{--}150 \text{ W}$ of heat from a chip area of a few cm^2 — heat fluxes comparable to a rocket nozzle. Without effective cooling, chips overheat within milliseconds. Solutions include heat sinks (extended surfaces to increase convective area), heat pipes (phase-change cooling), liquid cooling loops, and thermoelectric coolers. Thermal management is a limiting factor in computer performance: the “power wall” that halted clock speed

increases around 2004 was fundamentally a thermal problem.

Buildings require thermal design to maintain comfortable interior temperatures without excessive energy consumption. R-value is the thermal resistance per unit area of an insulating material ($\text{m}^2\cdot\text{K}/\text{W}$). Higher R-values mean better insulation. The total heat loss from a building is $\dot{Q} = A\Delta T/R_{\text{total}}$, where the sum includes walls, roof, windows, and air infiltration. Building energy codes specify minimum R-values to limit energy consumption.

Key Point: The Carnot limit and the laws of thermodynamics define the efficiency ceiling for any heat engine: no device can have efficiency greater than $1 - T_{\text{C}}/T_{\text{H}}$. Refrigerators and heat pumps obey analogous efficiency limits (COP limits). These are hard physical constraints that no engineering innovation can circumvent; they can only be approached, never exceeded.

53.9 Electricity, Circuits, and Control Systems

Electrical engineering translates the principles of electromagnetism (Part VI) into power systems, electronics, communications, and control. Every electrical system obeys Ohm's law ($V = IR$), Kirchhoff's laws (current and voltage conservation), and the principles of electromagnetic induction.

Power Systems

Electrical power is transmitted over long distances as high-voltage AC (alternating current) because transformers can step voltage up and down efficiently — high voltage means low current, which means low resistive losses ($P_{\text{loss}} = I^2R$). The national grid transmits at 100–500 kV; distribution to homes steps down to 230 V (UK/Europe) or 120 V (North America). The frequency is maintained at 50 or 60 Hz — deviating by even a fraction of a Hz causes equipment damage and grid instability.

Electronics

Transistors (Chapter 48) are the fundamental switching and amplifying element of all modern electronics. A smartphone contains approximately 15 billion transistors. Moore's Law — the observation that transistor density roughly doubles every two years — has driven computing performance for decades by making transistors smaller, faster, and cheaper. Quantum mechanical tunnelling, however, sets a fundamental limit to how small transistors can be made before electrons tunnel through the barrier that should switch them off.

Control Systems

A control system uses sensors, feedback, and actuators to maintain a desired output. A thermostat is a simple control system: it senses temperature, compares it to a setpoint, and switches the heater on or off. A cruise control system uses vehicle speed as the feedback signal and adjusts engine throttle to maintain target speed. Modern aircraft fly-by-wire systems replace mechanical linkages with electronic control systems: the pilot's stick inputs are processed by computers that calculate the control surface deflections needed to achieve the desired aircraft motion, checking hundreds of physical and safety constraints simultaneously.

Control theory uses the mathematics of feedback to analyse stability, response time, and precision. A poorly designed feedback system can oscillate or become unstable — a consequence directly analogous to mechanical resonance. The physics of the process being controlled determines what control strategies are feasible.

53.10 Waves, Sound, and Optical Technologies

Wave physics (Part V) underlies a wide range of engineering technologies in communications, sensing, imaging, and manufacturing.

Communications

Radio, microwave, cellular, WiFi, and satellite communications all use electromagnetic waves. The electromagnetic spectrum is a finite resource: different frequency bands are allocated for different uses and licensed by governments. Signal encoding, antenna design, and receiver sensitivity are all governed by electromagnetic wave physics and information theory. Fibre optic communications transmit data as light pulses through glass fibres using total internal reflection (Chapter 24), achieving data rates of terabits per second over transoceanic distances.

Ultrasonic Technology

Ultrasonic waves (sound above 20 kHz) are used for non-destructive testing (detecting cracks and voids in materials without cutting them open), sonar (ranging in water), medical imaging (ultrasound), and cleaning. The physics of reflection, transmission, and refraction at material interfaces (Chapter 22) determines what ultrasound can detect and with what resolution.

Optical Engineering

Cameras, telescopes, microscopes, laser systems, and optical fibres all apply the principles of geometric and wave optics (Chapters 23–25). The diffraction limit (Chapter 25) sets the minimum resolvable feature size for any optical instrument: smaller wavelengths achieve higher resolution. Electron microscopes use the de Broglie

wavelength of electrons (Chapter 45) to resolve features far smaller than visible light allows. Extreme ultraviolet lithography uses 13.5 nm wavelength EUV light to pattern transistors on computer chips.

RADAR and LIDAR

RADAR (Radio Detection And Ranging) uses the Doppler effect (Chapter 22) and reflection of radio waves to detect and locate objects, measure their speed, and map terrain. Weather RADAR detects precipitation by the reflection of radio waves from raindrops. Traffic speed cameras use the Doppler shift of reflected radio waves to measure vehicle speed. LIDAR (Light Detection And Ranging) uses laser pulses for high-resolution three-dimensional mapping and is a key sensor technology in autonomous vehicles.

53.11 Modelling, Prototyping, and Testing

Engineering design does not move directly from equations to manufactured product. Between theoretical analysis and production lies an iterative cycle of modelling, prototyping, testing, and refinement.

Mathematical and Computational Modelling

Engineers build mathematical models of proposed designs to predict performance before physical construction. Analytical models (closed-form equations) are used where the geometry is simple enough. Computational models — Finite Element Analysis (FEA) for stress and deformation, Computational Fluid Dynamics (CFD) for flow problems, circuit simulation for electronics — are used for complex geometries and coupled phenomena. All computational models are approximations: their accuracy depends on mesh quality, material model correctness, and boundary condition specification. Validating computational models against physical measurements is a critical engineering practice.

Prototyping

A prototype is a physical realisation of a design at some stage of development — from a rapid 3D-printed concept model to a full-scale functional prototype. Physical prototypes reveal interactions between components that are difficult to model computationally: assembly tolerances, unexpected resonances, thermal gradients, and manufacturing variability. Testing a prototype under realistic conditions is the most reliable way to identify design flaws before full production.

Testing and Measurement

Testing verifies that a product meets its specifications under real conditions. Mechanical testing measures strength, stiffness, fatigue life, and failure mode. Environmental testing exposes products to temperature extremes, humidity, vibration, and shock. Electrical testing checks performance across voltage and temperature ranges. Non-destructive testing (NDT) — X-ray, ultrasound, eddy current inspection — verifies internal quality without destroying the part. Every test is ultimately a physics experiment, and test results are meaningful only if the measurement uncertainty is understood and controlled.

53.12 Safety Factors and Reliability

Engineering design always includes margins to account for uncertainty. A safety factor (also called a factor of safety, FS) is the ratio of the material's strength (or a structure's load-carrying capacity) to the expected maximum applied stress (or load):

$$\text{FS} = \frac{\text{Strength}}{\text{Applied_stress}} \quad (\text{or Load capacity} / \text{Design load})$$

Factor of safety. FS = 1 means the structure will just barely not fail under the design load. Real designs use FS = 1.5 to 10 or more depending on the consequences of failure, uncertainty in loads and materials, and regulatory requirements.

Safety factors exist because of uncertainties: uncertainty in the actual loads (what if the maximum earthquake is larger than predicted?), uncertainty in material properties (batch-to-batch variability, effects of corrosion and aging), uncertainty in analysis methods (simplified models miss some failure modes), and uncertainty in construction quality (workmanship defects).

Reliability engineering quantifies the probability that a system will perform its function over a specified time period. Probabilistic analysis replaces single-number safety factors with distributions of load and strength, calculating the probability of failure rather than assuming failure occurs only when the mean load exceeds the mean strength. For high-consequence applications (aircraft, nuclear plants, life-critical medical devices), failure probabilities as low as 10^{-9} per hour of operation are required.

Application	Typical Factor	Safety	Reason for Level
Aircraft primary structure	1.5		Weight penalty is severe; materials are well-characterised
Automotive structural	2–3		Moderate weight concern; load

		variability is large
Building structural steel	1.67 (AISC LRFD)	Well-characterised loads; inspectable; code-mandated
Pressure vessels (ASME)	3.5 (on UTS)	Burst failure is catastrophic; weld defects increase risk
Crane hooks and lifting gear	5–10	Dynamic loads; life-safety consequence of failure
Ropes, climbing equipment	10–15	Shock loads; user life at stake; degradation in service

Table 53.3 *Typical safety factors in various engineering applications and the rationale for their selection.*

Common Mistake: A safety factor does not make a design infinitely safe — it reduces the probability of failure to an acceptable level. An unexpected failure mode not covered by the analysis (fatigue, corrosion, an unforeseen load combination) can cause failure even at high safety factors. Engineering safety requires identifying all relevant failure modes, not just calculating a number.

53.13 Common Mistakes in Applying Physics to Engineering

- Treating idealised models as exact descriptions: Assuming a structure is rigid, a fluid is inviscid, or a gas is ideal when the real system deviates significantly from the idealisation. Always check whether the model's assumptions are satisfied in the actual application.
- Ignoring units and dimensional consistency: Mixing units (pounds and newtons, inches and metres, gauge and absolute pressure) causes catastrophic errors. The 1999 Mars Climate Orbiter was lost because one engineering team used pound-force-seconds and another used newton-seconds for impulse — a \$327 million unit conversion error.
- Neglecting dynamic loads and fatigue: Static analysis finds the stress under steady loading. But real structures experience fluctuating, dynamic, and impact loads that can cause fatigue failure at stresses well below the static yield strength. A design that is safe statically can still fail in service if fatigue is not analysed.
- Confusing thermal and electrical analogues incorrectly: Thermal and electrical systems have analogous mathematics (Fourier's law mirrors Ohm's law; thermal resistance mirrors electrical resistance), but they differ in important ways. Heat

capacity has no perfect electrical analogue, and thermal interfaces have contact resistance that electrical contacts do not.

- Applying Bernoulli's equation outside its domain: Bernoulli's equation applies to steady, inviscid, incompressible flow along a streamline. Applying it to highly viscous flows, compressible flows (high-speed air), or flows with large velocity gradients gives incorrect results. Know the limits of every equation.
- Over-relying on safety factors without identifying failure modes: A safety factor is only as good as the analysis that produced it. If the failure mode is wrong (using tensile yield strength when buckling is the actual mode, for example), the safety factor is meaningless. Failure mode analysis must precede safety factor application.
- Ignoring manufacturing tolerances and variability: A design that works perfectly in analysis may fail in production because parts are not made to exact dimensions. Engineers specify tolerances and verify that performance is acceptable across the entire tolerance range, not just at the nominal (target) dimension.

Historical Note: The Challenger space shuttle disaster of 1986 was caused by O-ring seals that became stiff and failed to seal properly at the low temperature (approximately -2°C) at launch. Engineers at the O-ring manufacturer had warned against launching at such temperatures, but organisational pressure overrode the technical objection. The disaster illustrates that engineering safety requires both correct physics and the institutional culture to act on technical concerns without organisational pressure to ignore them.

Chapter Summary

Engineering is physics translated into purpose. It uses the laws of mechanics, thermodynamics, electromagnetism, fluids, and waves to design systems that must work safely and reliably under real-world constraints.

- Engineering design is constrained by physical laws (non-negotiable) and practical requirements (tradeoffs). Improving one parameter often worsens another.
- Structural engineering applies statics, stress analysis ($\sigma = E\epsilon$), and failure mode analysis (yielding, fracture, fatigue, buckling) to design safe load-bearing systems.
- Machine design applies Newton's laws, torque, gear ratios, and power conservation to convert and transmit force and motion.
- Thermal engineering applies Fourier's law, convection, radiation, and the laws of thermodynamics (Carnot efficiency) to design systems that manage heat flows.

- Hydraulic and aerodynamic systems apply Bernoulli's equation, continuity, and momentum principles to transmit power and generate lift.
- Electrical engineering applies Ohm's law, Kirchhoff's laws, and electromagnetic induction to power systems, electronics, and control.
- Wave physics underlies communications (electromagnetic waves), imaging (optics, ultrasound), and sensing (RADAR, LIDAR, sonar).
- Modelling, prototyping, and testing form an iterative cycle that moves a design from equations to a product that performs as intended.
- Safety factors account for uncertainty in loads, materials, and analysis. Reliability engineering quantifies failure probability for high-consequence applications.
- Common engineering errors include unit confusion, ignoring dynamic loads and fatigue, applying models outside their validity range, and not identifying all failure modes before assigning a safety factor.

Key Terms

dead load: A permanent, static load on a structure, such as the weight of the structure itself and any permanently attached fixtures.

live load: A variable or movable load, such as occupants, vehicles, wind, or snow.

stress (σ): Force per unit area within a material; SI unit Pa. Normal stress acts perpendicular to a surface; shear stress acts parallel.

strain (ϵ): Fractional deformation; $\epsilon = \Delta L/L_0$; dimensionless. Elastic strain is recovered when load is removed; plastic strain is permanent.

Young's modulus (E): The ratio of stress to strain in the elastic (linear) range; $E = \sigma/\epsilon$; units Pa. Measures material stiffness.

yield strength: The stress at which a material begins to deform plastically (permanently). Engineering design keeps stresses below yield strength.

fatigue: Failure due to cyclic loading at stresses below the static yield strength; results from slow crack growth under repeated stress cycles.

buckling: A geometric instability failure mode for slender columns under compression; occurs below the material's yield strength.

factor of safety (FS): The ratio of a material's strength to the applied stress; $FS > 1$ provides margin against uncertainty and variability.

efficiency (η): The ratio of useful energy output to total energy input; $\eta = W_{\text{out}}/W_{\text{in}}$; always ≤ 1 for real machines.

Carnot efficiency: $\eta_{\text{Carnot}} = 1 - T_{\text{C}}/T_{\text{H}}$; the maximum possible efficiency of a heat engine operating between temperatures T_{H} and T_{C} .

mechanical advantage (MA): The ratio of output force to input force in a machine; $\text{MA} > 1$ amplifies force at the cost of output distance.

gear ratio (N): The ratio of output to input tooth count in a gear pair; torque is multiplied by N, angular velocity divided by N.

thermal resistance (R_{th}): $R_{\text{th}} = \Delta T / \dot{Q}$; analogous to electrical resistance; allows complex thermal systems to be analysed as circuit-like networks.

finite element analysis (FEA): A numerical method for calculating stress, deformation, and other field quantities in complex geometries by dividing the structure into small elements.

non-destructive testing (NDT): Inspection techniques (ultrasound, X-ray, eddy current) that verify internal quality without damaging the part.

reliability: The probability that a system performs its required function over a specified period under specified conditions; quantified in reliability engineering.

control system: A system that uses sensors, feedback, and actuators to maintain a desired output by comparing actual to desired performance and correcting the difference.

Concept Review Questions

Answer in complete sentences.

1. Explain the difference between a physical constraint and a practical constraint in engineering design. Give one example of each for the design of a passenger aircraft.
2. A beam under a transverse load experiences both tension and compression. Explain where in the cross-section these stresses occur and why. Which material property is most important for resisting each?
3. Explain why fatigue failure can occur at stresses well below the material's static yield strength. Why is this failure mode particularly dangerous?
4. A hydraulic car lift and a mechanical lever both provide force amplification. Explain the physical principle underlying each. What do they have in common?
5. Explain why electrical power is transmitted at high voltage and low current over long distances. What physics principle determines the energy loss in the transmission line?

6. A design engineer specifies a safety factor of 5 for a crane hook. What uncertainties does this safety factor account for? Does a safety factor of 5 guarantee the hook will never fail? Explain.
 7. Explain why increasing the operating temperature of a heat engine improves its efficiency. What material challenge does this create, and how do engineers address it?
 8. List three engineering applications of wave physics and identify the specific wave physics principle each exploits (e.g., reflection, Doppler effect, diffraction).
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Engineering Design Analysis

9. A steel cable ($\sigma_{\text{yield}} = 400 \text{ MPa}$, $E = 200 \text{ GPa}$) supports a load of 120 kN. (a) What minimum cross-sectional area is required for a safety factor of 3 against yielding? (b) What is the diameter of a circular cable with this area? (c) If the cable is 20 m long, how much does it stretch under load?
 10. A concrete column ($E = 30 \text{ GPa}$, $\sigma_{\text{compressive strength}} = 30 \text{ MPa}$) is 4 m tall and has a $0.25 \text{ m} \times 0.25 \text{ m}$ cross-section. (a) What is its cross-sectional area? (b) What maximum load can it carry at a safety factor of 4? (c) By how much does it compress under this load?
 11. A hollow aluminium tube (outer diameter 60 mm, inner diameter 50 mm, $E = 70 \text{ GPa}$) is used as a column of length 2.0 m. Calculate (a) the cross-sectional area and (b) the stress if it carries a compressive load of 80 kN. Is this below the yield strength of aluminium (270 MPa)?
 12. A bolted joint uses four steel bolts, each with a cross-sectional area of 80 mm^2 , to carry a shear load of 100 kN. (a) Calculate the shear stress on each bolt. (b) If the shear yield strength of the steel is 240 MPa and the safety factor required is 2, are the bolts adequate?
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Applied Physics Case Studies

13. A steam turbine power plant operates with steam at $T_{\text{H}} = 550^\circ\text{C}$ and condenses at $T_{\text{C}} = 45^\circ\text{C}$. (a) Calculate the Carnot efficiency. (b) If the actual thermal efficiency is 38%, what fraction of the Carnot efficiency is achieved? (c)

The plant generates 500 MW of electrical power. At 38% efficiency, at what rate is heat rejected to the cooling water?

14. A pump delivers $0.08 \text{ m}^3/\text{s}$ of water through a pipe system. The total head required (elevation + friction) is 40 m. (a) Calculate the ideal hydraulic power ($P = \rho g Q h$). (b) If the pump has an efficiency of 72%, what is the electrical power input? (c) If electricity costs £0.20 per kWh, what is the hourly operating cost?
15. A gear reducer takes an input of 1500 rpm and 50 N·m of torque and produces an output of 300 rpm. (a) What is the gear ratio N ? (b) What is the output torque (assume ideal, no friction losses)? (c) Calculate input and output power and verify they are equal.
16. A building wall section consists of: 10 mm plasterboard ($R = 0.06 \text{ m}^2\text{K/W}$), 100 mm mineral wool insulation ($R = 2.50 \text{ m}^2\text{K/W}$), 100 mm brick ($R = 0.12 \text{ m}^2\text{K/W}$). Interior temperature = 20°C , exterior = -5°C . (a) Calculate the total thermal resistance R_{total} . (b) Calculate the heat flux (W/m^2) through the wall. (c) For a wall area of 15 m^2 , what is the total heat loss rate in watts?

Failure Analysis Exercises

17. A steel shaft (diameter 30 mm, fatigue limit 200 MPa) rotates at 3000 rpm and carries a bending load that produces a cyclic stress of 180 MPa. (a) Is the shaft below its fatigue limit? (b) How many stress cycles occur per second? (c) If the fatigue limit is instead 160 MPa (a different steel grade), what happens over time?
18. A slender steel column (length 3 m, cross-sectional area 5 cm^2 , radius of gyration 15 mm) buckles at an Euler critical load $P_{\text{cr}} = \pi^2 EI/L^2$, where $E = 200 \text{ GPa}$ and $I = Ar^2$. (a) Calculate the second moment of area I . (b) Calculate the Euler critical load P_{cr} . (c) If the applied compressive load is 80 kN, is buckling a concern?
19. During a quality inspection, an ultrasonic test detects a crack of length 5 mm in a steel pressure vessel wall. The fracture toughness of the steel is $K_{\text{Ic}} = 50 \text{ MPa}\sqrt{\text{m}}$. The operating stress is 120 MPa. Using $K = \sigma\sqrt{\pi a}$ (simplified fracture mechanics), where a is the half-crack length, calculate the stress intensity factor K and compare it to K_{Ic} . Should the vessel be taken out of service?

Design Tradeoff Problems

20. You are designing a bicycle frame. Consider three materials: steel (density 7870 kg/m^3 , yield strength 500 MPa), aluminium alloy (density 2700 kg/m^3 , yield strength 270 MPa), and carbon fibre composite (density 1600 kg/m^3 , tensile strength 600 MPa). (a) Calculate the specific strength (yield strength / density) for each material. (b) Which material gives the lightest frame for the same strength? (c) What other factors should a designer consider beyond specific strength?
21. A data centre cooling system can use: (a) air cooling (simple, low cost, limited to 2 kW per rack); (b) liquid cooling (moderate cost, can handle 40 kW per rack); (c) immersion cooling (highest cost, can handle 200 kW per rack). High-performance AI computing requires 80 kW per rack. Analyse the tradeoffs and justify which cooling method you would specify.
22. A new bridge is to be built across a 300 m wide estuary. Available options include: (a) a beam bridge (maximum span $\sim 50 \text{ m}$ without intermediate piers); (b) a cable-stayed bridge (spans $200\text{--}500 \text{ m}$); (c) a suspension bridge (spans $500 \text{ m--}3000 \text{ m}$). Which type is most appropriate? What physical constraints drive this choice? What additional information would you need to finalise the selection?
23. An autonomous vehicle requires a sensor for detecting obstacles up to 200 m away in all weather conditions, day and night. Compare RADAR (radio waves, $\lambda \sim \text{mm}$) and LIDAR (laser, $\lambda \sim 900 \text{ nm}$) on the basis of: range resolution (related to wavelength), angular resolution (diffraction limit $\approx \lambda/D$ for aperture D), weather performance, and cost. Which sensor or combination would you recommend and why?

Chapter 53 Checkpoint

Before moving on, confirm you can do each of the following.

I can...	See Section
Map engineering disciplines to their foundational physics areas	53.1
Explain design constraints and tradeoffs with an example	53.2
Calculate stress and strain and identify structural failure modes	53.3–53.4

Apply energy conservation and Carnot efficiency to thermal machines	53.5
Analyse gear systems using torque, speed, and power relationships	53.6
Explain hydraulic, aerodynamic, and pipe flow engineering applications	53.7
Apply thermal resistance concepts to heat transfer engineering problems	53.8
Explain power transmission, transistor switching, and control feedback	53.9
Identify engineering applications of wave reflection, Doppler, and diffraction	53.10
Describe the role of modelling, prototyping, and testing in engineering design	53.11
Calculate and interpret safety factors; explain reliability engineering	53.12
Identify and explain common errors in applying physics to engineering	53.13

Continue to Chapter 54: Medical and Biological Physics →